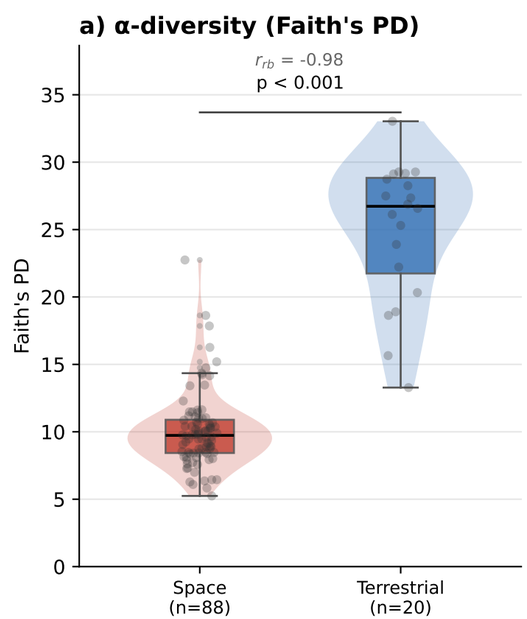
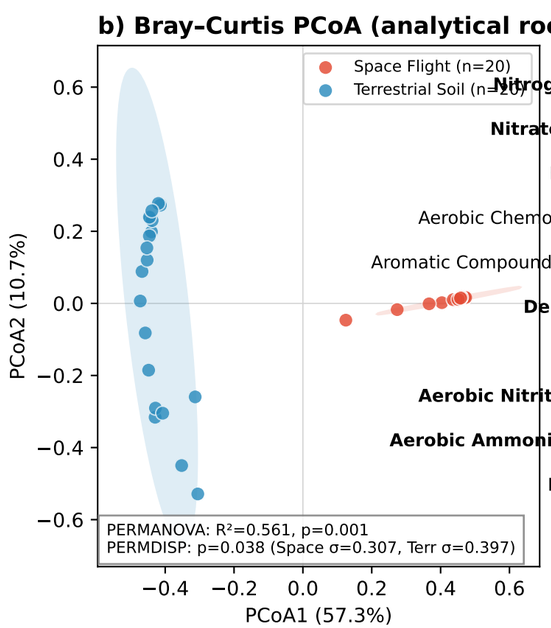
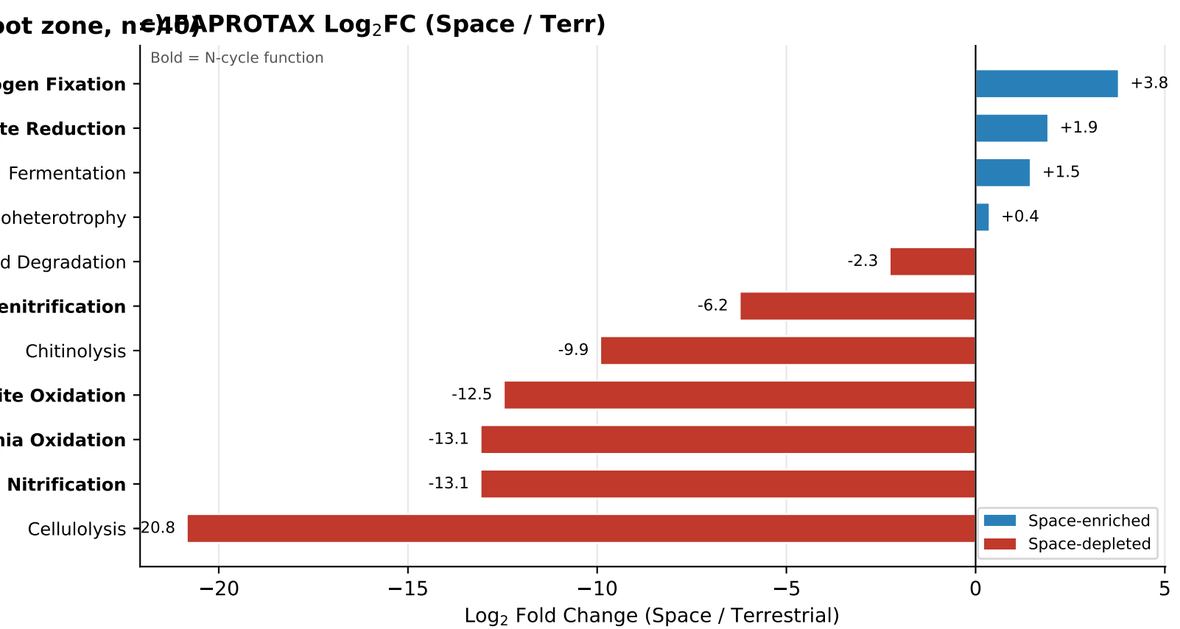
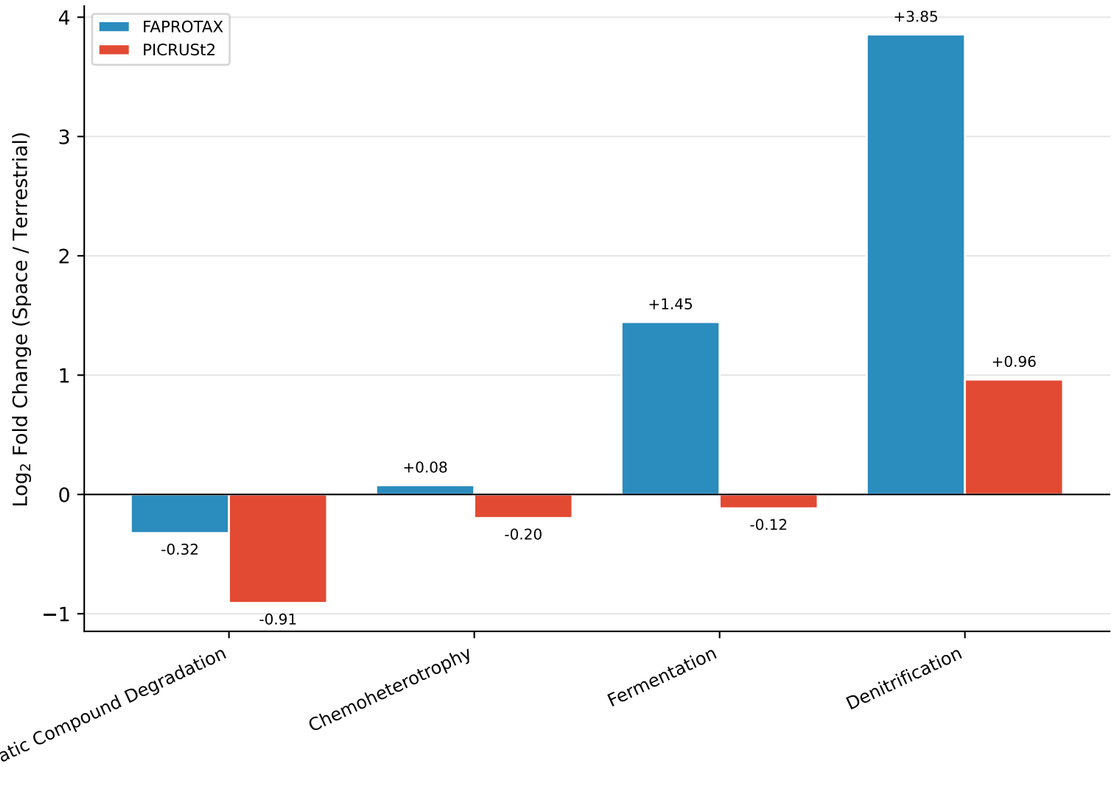
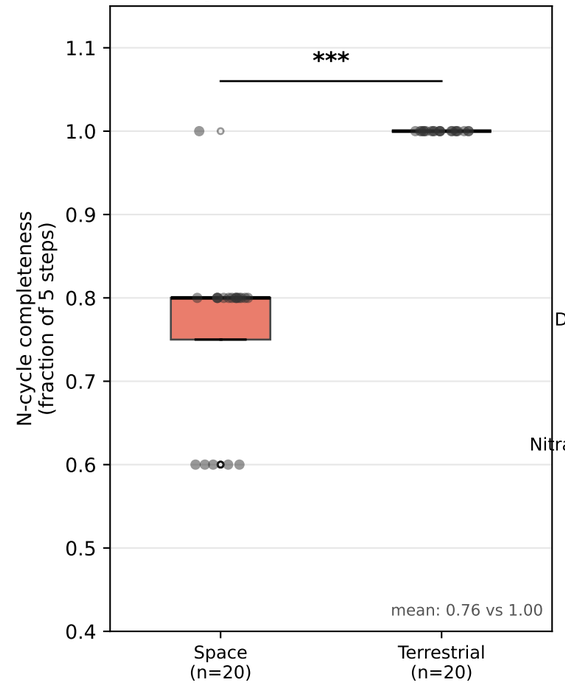
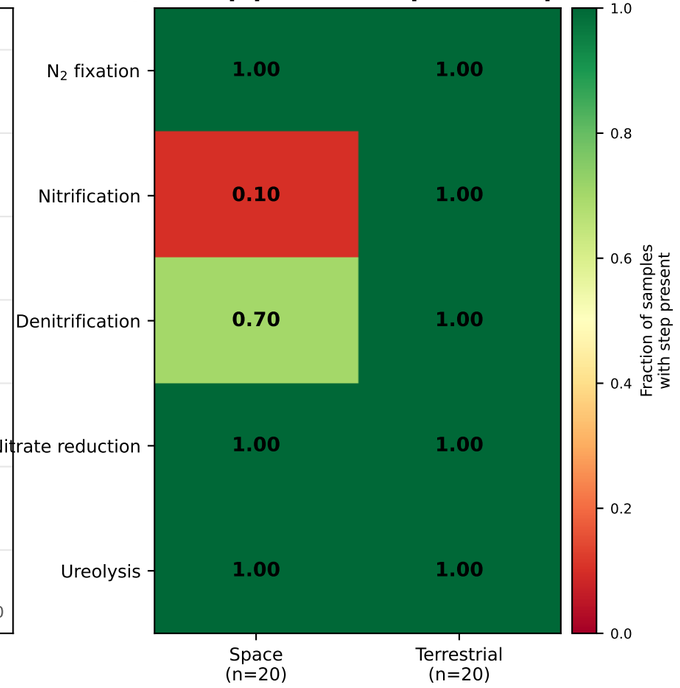

a) α-diversity (Faith's PD)
r_{rb} = -0.98
p < 0.001
Faith's PD
Space
(n=88)
Terrestrial
(n=20)


b) Bray–Curtis PCoA (analytical root zone, n=108)
Space Flight (n=20)
Terrestrial Soil (n=20)
PERMANOVA: R²=0.561, p=0.001
PERMDISP: p=0.038 (Space σ=0.307, Terr σ=0.397)
PCoA1 (57.3%)
PCoA2 (10.7%)


c) FAPROTAX Log2FC (Space / Terr)
Bold = N-cycle function
Nitrogen Fixation
Nitrate Reduction
Fermentation
Aerobic Chemoheterotrophy
Aromatic Compound Degradation
Denitrification
Chitinolysis
Aerobic Nitrite Oxidation
Aerobic Ammonia Oxidation
Nitrification
Cellulolysis
+3.8
+1.9
+1.5
+0.4
-2.3
-6.2
-9.9
-12.5
-13.1
-13.1
-20.8
Space-enriched
Space-depleted
Log2 Fold Change (Space / Terrestrial)


d) FAPROTAX × PICRUSt2 consensus (MetaCyc-covered functions)
FAPROTAX
PICRUSt2
Log2 Fold Change (Space / Terrestrial)
+0.08
+1.45
+3.85
-0.32
-0.91
-0.20
-0.12
+0.96
Aromatic Compound Degradation
Chemoheterotrophy
Fermentation
Denitrification


e) Completeness per sample (0.76 vs 1.00)
***
N-cycle completeness
(fraction of 5 steps)
mean: 0.76 vs 1.00
Space
(n=20)
Terrestrial
(n=20)


f) Step presence (Space collapses on nitrification)
N2 fixation
Nitrification
Denitrification
Nitrate reduction
Ureolysis
1.00
1.00
0.10
1.00
0.70
1.00
1.00
1.00
1.00
1.00
Fraction of samples
with step present
Space
(n=20)
Terrestrial
(n=20)